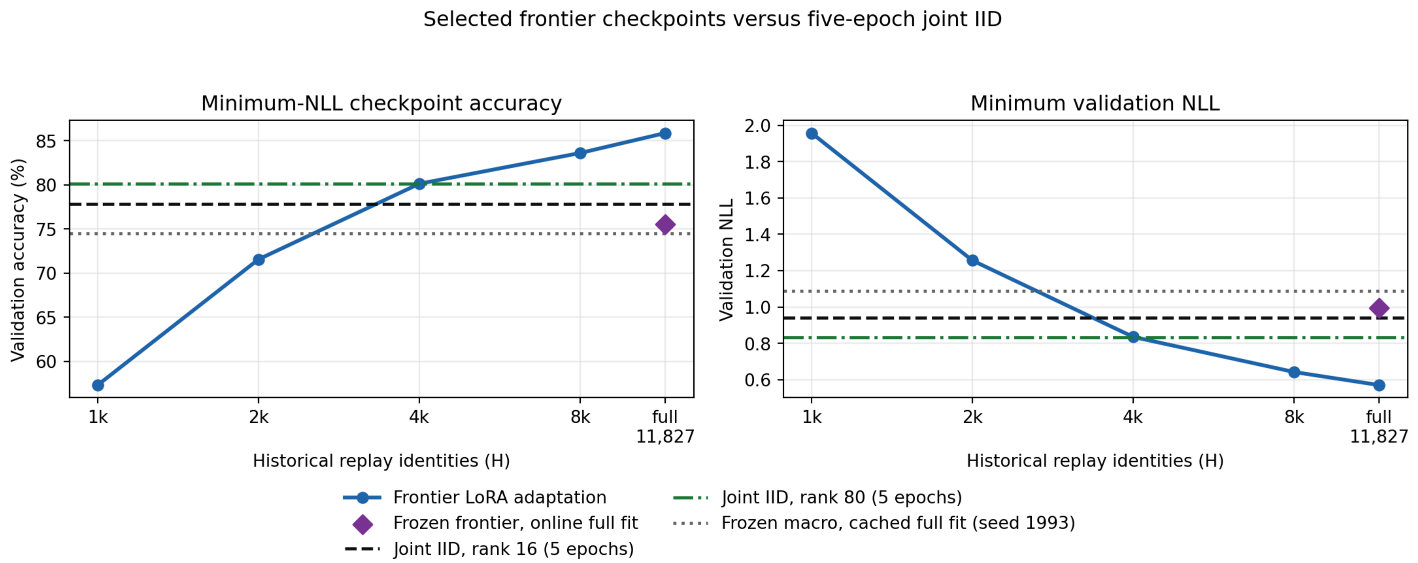
ImageNet-R stage-31 frontier-LoRA adaptation

Can jointly adapting five fragmented node representations close the same-split joint-IID accuracy and NLL gap?

Selected-checkpoint result. The minimum-NLL adaptive condition is Frontier LoRA adaptation (H=11,827; full fit): **85.864% accuracy / 0.5710 NLL**. That is 8.003 percentage points higher and 0.3715 NLL lower than the five-epoch rank-16 joint-IID reference. Frontier LoRA adaptation (H=4,096) is the smallest H to reach both rank-16 joint-IID metrics, first doing so at epoch 6.

Aggregate-rank result. Rank-80 joint IID reaches **80.125% / 0.8339 NLL** at epoch five. The larger adapter recovers 32.2% of the frontier's accuracy advantage and 30.8% of its NLL advantage over rank-16 joint IID.



Blue frontier points use each condition's minimum-NLL checkpoint. Black rank-16 and green rank-80 joint-IID lines are fixed epoch-five endpoints on the identical fit and validation identities.

Aggregate-rank comparison

Condition	LoRA layout	LoRA parameters	Other trainable parameters	Epoch-5 accuracy	Epoch-5 NLL
Joint IID, rank 16 (5 epochs)	1 x 16	1,327,104	95,356	77.862%	0.9425
Joint IID, rank 80 (5 epochs)	1 x 80	6,635,520	95,356	80.125%	0.8339
Adaptive frontier, full history (epoch 5)	5 x 16	6,635,520	12,055,496	84.880%	0.5897

All three rows use exactly five complete passes over the 12,194-image fit population. “Other” means the joint classifier or the frontier macro integrator; frozen base-model parameters are omitted.

Complete condition summary

Condition	Min-NLL epoch	Accuracy at min NLL	Minimum NLL	Maximum accuracy	NLL at max accuracy	Runtime
Frontier LoRA adaptation (H=1,024)	31	57.330%	1.9572	57.527%	1.9624	21.9 min
Frontier LoRA adaptation (H=2,048)	25	71.564%	1.2565	71.564%	1.2565	30.1 min
Frontier LoRA adaptation (H=4,096)	12	80.125%	0.8360	80.846%	0.8443	47.1 min
Frontier LoRA adaptation (H=8,192)	8	83.601%	0.6430	84.093%	0.6534	88.2 min
Frontier LoRA adaptation (H=11,827; full fit)	13	85.864%	0.5710	85.864%	0.5710	120.0 min
Frozen frontier, online full fit	16	75.533%	0.9967	76.418%	1.0063	43.4 min

Maximum accuracy and its NLL are reported separately to expose calibration tradeoffs. Every cell also includes all 367 current-task identities; maximum historical H=11,827 is exactly the 12,194-image full fit.

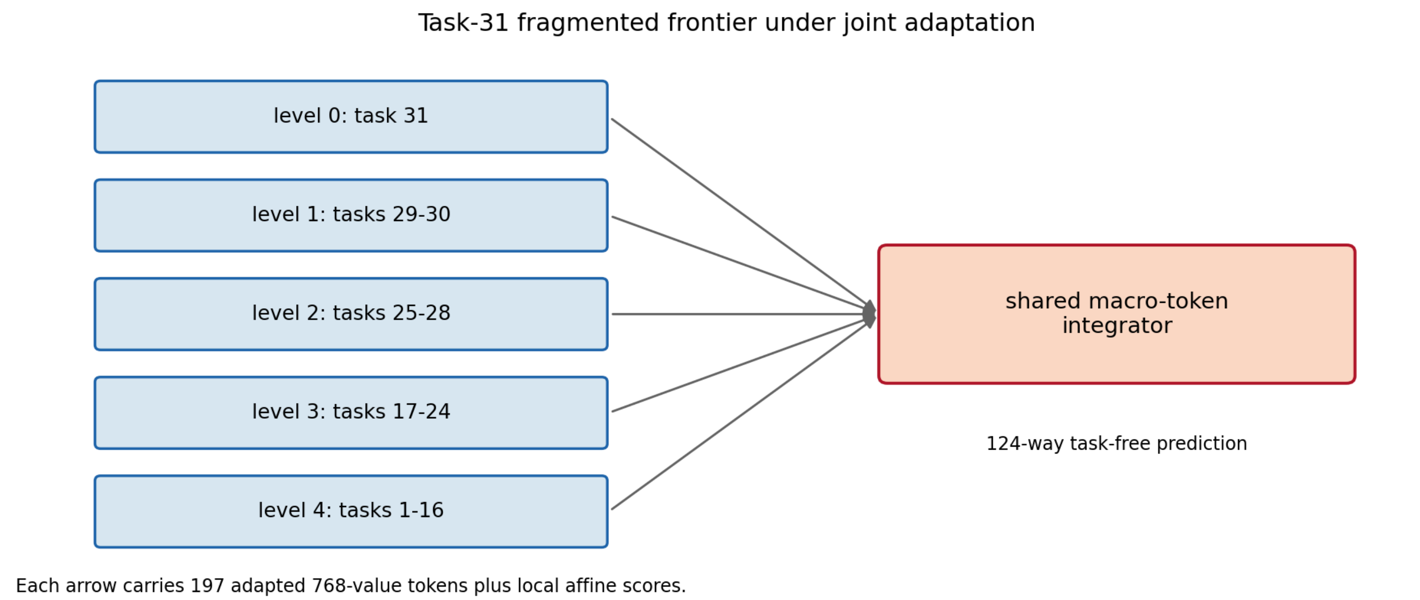
Rank-80 five-epoch trajectory

Epoch	Train objective accuracy	Train objective NLL	Validation accuracy	Validation NLL
1	50.672%	2.3264	75.992%	0.9977
2	72.593%	1.1689	78.485%	0.8350
3	77.358%	0.9351	80.026%	0.7954
4	80.007%	0.8119	80.354%	0.8340
5	82.221%	0.7234	80.125%	0.8339

Epoch five is the predeclared comparison endpoint. The minimum validation NLL occurs at epoch three; it is retained only as a convergence diagnostic.

Architecture and experimental boundary

The task-31 frontier has five nodes at levels 0-4, covering task intervals 31, 29-30, 25-28, 17-24, and 1-16. Each node supplies its own final 197 x 768 LoRA-adapted token sequence and immutable local affine scores. The macro transformer combines them without task IDs or labels.



The base ViT and node classifiers are frozen. Only five rank-16 LoRAs and the shared 12.06M-parameter macro head can move.

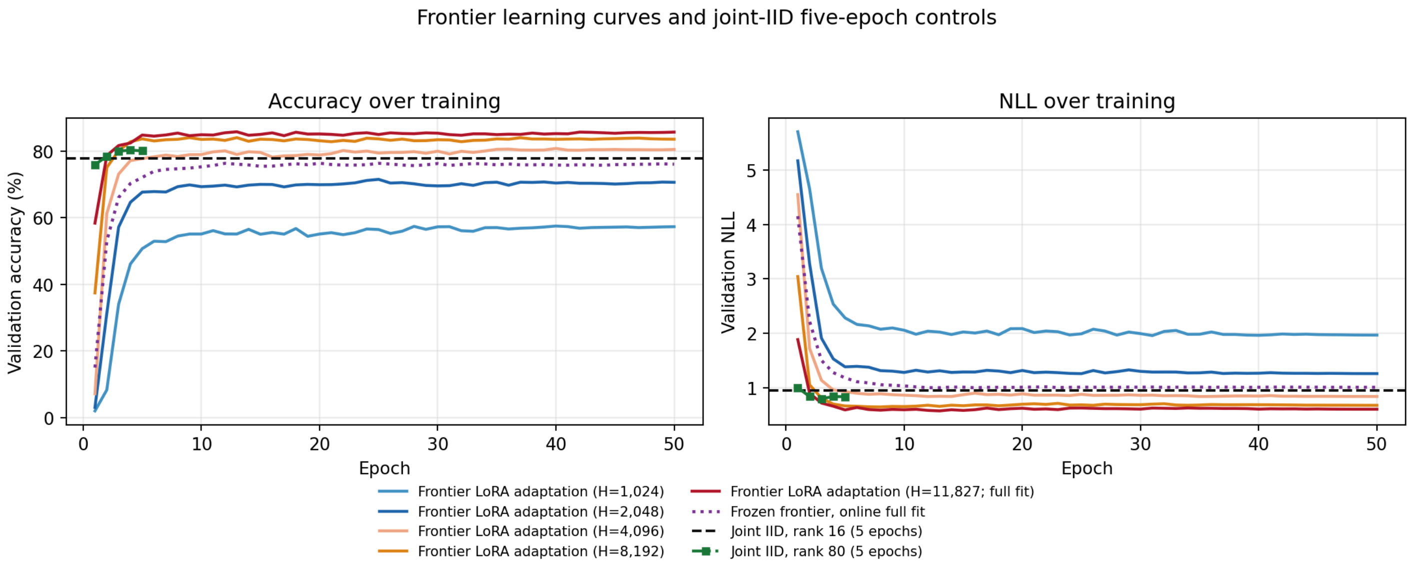
Every cell includes all 367 current-task images. H=1,024, 2,048, 4,096, 8,192, and 11,827 are nested prefixes of one deterministic uniform draw from the historical tasks without replacement. Maximum H therefore gives exactly all 12,194 fit identities. Every cell starts independently from identical sealed node tensors and macro initialization. The validation partition has 3,049 clean identities and never contributes gradients. No test image is opened.

Compute boundary. The adaptive frontier evaluates five node-specific ViTs and the macro transformer for every image. Joint IID evaluates one ViT with one shared LoRA and classifier. Their split and full-fit data exposure match, but parameter count, training compute, and deployment compute do not.

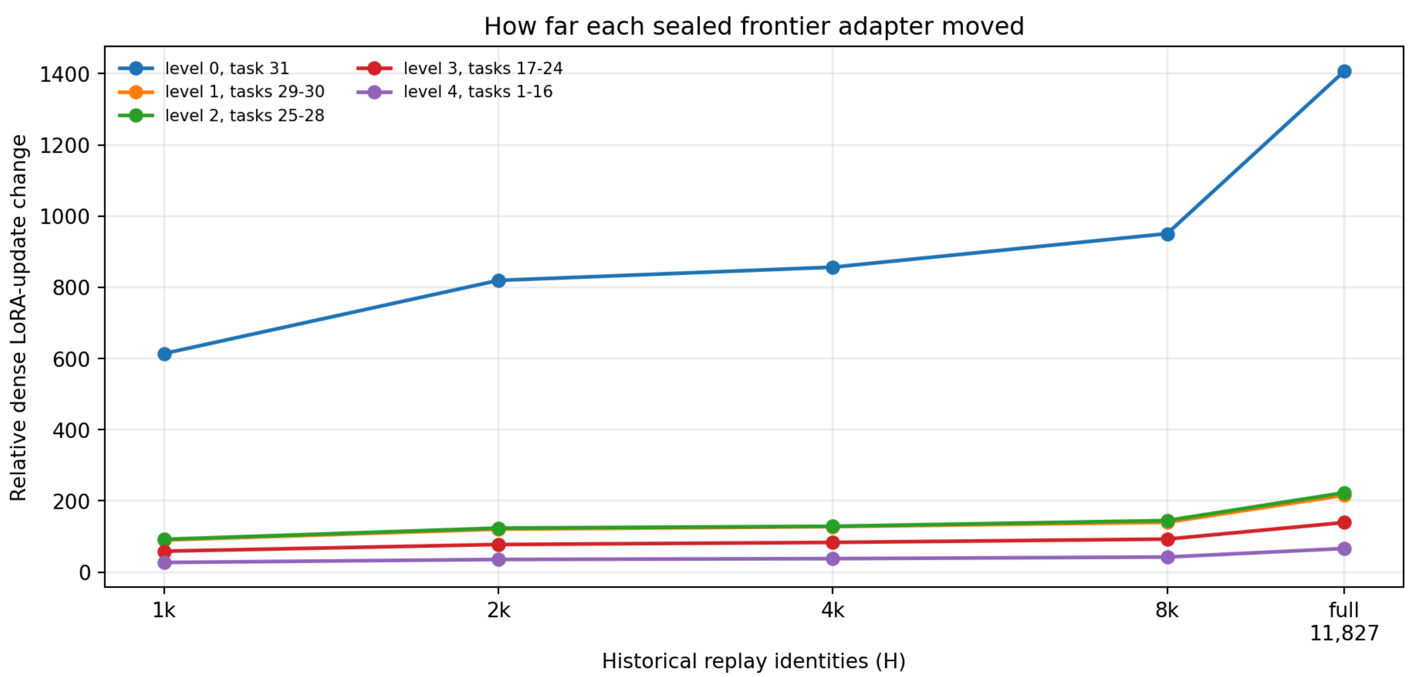
Why the frozen online control exists

The previous frozen macro was trained from cached center-crop tokens. This run loads images online with deterministic random training augmentation. The purple full-fit control follows that new path while freezing the LoRAs, so its difference from the cached gray reference measures the pipeline/augmentation change rather than representation adaptation.

Optimization behavior



Every frontier epoch and all five rank-80 joint-IID epochs are retained in hash-chained histories. The dashed black line is the rank-16 epoch-five endpoint, not a stopping gate.



Scale-aware Frobenius movement of each dense LoRA update, measured from its sealed source node at the selected minimum-NLL checkpoint.

Interpretation

Allowing the frontier LoRAs to move changes the result by 10.331 percentage points and 0.4257 NLL relative to the online frozen-LoRA control at their minimum-NLL checkpoints. H=4,096 is the smallest tested historical population to cross both rank-16 joint-IID values. At exactly 5 full-fit passes (60,970 image presentations), adaptive full history is 84.880% / 0.5897, frozen full history is 72.155% / 1.1788, and rank-16 joint IID is 77.862% / 0.9425. This isolates feature adaptation from training exposure, but not the frontier's extra model and deployment compute.

What the rank match does—and does not—show

The new rank-80 joint-IID control reaches **80.125% accuracy / 0.8339 NLL** at the fixed epoch-five endpoint. Increasing the joint adapter from rank 16 to rank 80 raises accuracy by 2.263 percentage points and lowers NLL by 0.1085. This recovers 32.2% of the frontier's accuracy advantage and 30.8% of its NLL advantage over rank-16 joint IID. The frontier remains 4.756 points higher and 0.2443 NLL lower.

Both sides expose exactly 6,635,520 trainable LoRA parameters and 60,970 image presentations. That is the full extent of the match. The adaptive frontier also trains a 12,055,496-parameter macro integrator, starts from five separately pretrained rank-16 adapters, and executes five ViT paths. Rank-80 joint IID starts one adapter from the standard zero-effect initialization, trains a 95,356-parameter classifier, and executes one ViT path. Its diagnostic minimum NLL is 0.7954 with 80.026% accuracy at epoch 3.

The head schedule is the previous minimum-NLL winner: effective batch 64, peak AdamW LR 3e-5, 50 epochs, 5% warmup, and cosine decay. The LoRA peak LR 5e-4 is imported from the same-split joint recipe but run here in AdamW under the shared schedule. It has not been tuned for this coupled model.

Limits. This is a one-seed validation screen. The H cells change both unique data and total optimizer updates. Maximum accuracy is exploratory because all 50 validation checkpoints are visible. A promising condition needs replication and a focused LoRA/head learning-rate audit before any locked-test use.

Integrity and reuse

The training seals record zero fit-validation overlap, zero test overlap, zero test evaluations, and unchanged source hierarchy files. Fresh-process replay authenticated all six frontier cells and the rank-80 control without taking an optimizer step. Large checkpoints and model weights remain local; compact histories, tables, plots, and protocol records are the report surface.